What Is React?

React is a JavaScript client-side framework that allows you to enhance HTML markup with custom JavaScript to allow for complex user interactions.[35] React uses a mini-language called JSX to allow you to write HTML-style markup inside your JavaScript or TypeScript code.

React is designed using a fundamentally different model of interaction than Stimulus. Stimulus is designed to allow you to add interactivity to HTML markup that you are already writing or have already written. In React, the markup is written using React's own JSX language, and React completely controls what is written to the DOM within its boundaries.

The two tools also handle internal state differently. In Stimulus, we keep state using the DOM itself or on the server and are responsible for updating the page when state changes. In React, state is kept within the React system. When state changes, the React run-time loop updates the DOM automatically to reflect the new state.

React's approach to displaying values is called *declarative* or *reactive*. Both terms refer to the same process. When it comes to displaying information in React, it is the developer's job to declare in code what the output should look like and the system's job to react to changes in values by redrawing the page.

React manages this process by maintaining its own virtual DOM and using that virtual DOM to update only the parts of the actual DOM that change when a state value changes. However, in order to allow the DOM to automatically change, React needs to be made aware when you change a value that is relevant to the display in the browser. Because of this, you can only change state in React using specific functions provided by the framework, which trigger the resulting browser display changes. As we'll see, React allows you to declare a specific value as being part of the changeable state, and when you do so, React provides you with a custom setter function that changes that value. When you use these setter functions, React notices the state change and updates the parts of the DOM